

LAB -1 (ASSIGNMENT)

1. Create a LabFiles directory and perform create/copy/move/rename/delete operations using touch, cp, mv, rm, unlink, ls, stat.

```

Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@tamanna:~$ mkdir Labfiles
mkdir: Labfiles: File exists
vboxuser@tamanna:~$ cd Labfiles
vboxuser@tamanna:~/Labfiles$ touch file1.txt file2.txt file3.txt
vboxuser@tamanna:~/Labfiles$ ls
file.txt  file1.txt  file2.txt  file3.txt
vboxuser@tamanna:~/Labfiles$ cp file1.txt copy_file1.txt
vboxuser@tamanna:~/Labfiles$ ls
copy_file1.txt  file.txt  file1.txt  file2.txt  file3.txt
vboxuser@tamanna:~/Labfiles$ mv file2.txt moved_file2.txt
vboxuser@tamanna:~/Labfiles$ ls
copy_file1.txt  file.txt  file1.txt  file3.txt  moved_file2.txt
vboxuser@tamanna:~/Labfiles$ mv file3.txt renamed_file3.txt
vboxuser@tamanna:~/Labfiles$ ls
copy_file1.txt  file.txt  file1.txt  moved_file2.txt  renamed_file3.txt
vboxuser@tamanna:~/Labfiles$ stat renamed_file3.txt
  File: renamed_file3.txt
  size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 8,2      Inode: 656173      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/vboxuser)   Gid: ( 1000/vboxuser)
Access: 2026-09-16 15:19:23.490875174 +0000
Modify: 2026-09-16 15:19:23.490875174 +0000
Change: 2026-09-16 15:21:38.157167382 +0000
 Birth: 2026-09-16 15:19:23.490875174 +0000
vboxuser@tamanna:~/Labfiles$ rm copy_file1.txt
vboxuser@tamanna:~/Labfiles$ ls
file.txt  file1.txt  moved_file2.txt  renamed_file3.txt
vboxuser@tamanna:~/Labfiles$ unlink moved_file2.txt
vboxuser@tamanna:~/Labfiles$ ls
file.txt  file1.txt  renamed_file3.txt
vboxuser@tamanna:~/Labfiles$ ls -l
total 0
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 16 15:06 file.txt
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 16 15:19 file1.txt
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 16 15:19 renamed_file3.txt
vboxuser@tamanna:~/Labfiles$
```

2. Demonstrate pipes & redirection (>, >>, , 2>>, 2>&1, |) with cat, grep, sort, uniq, wc, tee to generate filtered reports.

```

Ubuntu 26.04.1 LTS tamanna tty6
tamanna login: vboxuser
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@tamanna:~$ cat > students.txt
Chikki
Tamanna
Tishu
Geetanjali
Rinki
Enu
Mihru
vboxuser@tamanna:~$ cat students.txt > output.txt
vboxuser@tamanna:~$ cat output.txt
Chikki
Tamanna
Tishu
Geetanjali
Rinki
Enu
Mihru
vboxuser@tamanna:~$ echo "Rahul" >> output.txt
vboxuser@tamanna:~$ wc -l < students.txt
7
vboxuser@tamanna:~$ cat students.txt | sort
Chikki
Enu
Geetanjali
Mihru
Rinki
Tamanna
Tishu
vboxuser@tamanna:~$ cat students.txt | grep "Chikki"
Chikki
vboxuser@tamanna:~$
```

```

1 Chikki
1 Enu
1 Geetanjali
1 Mihru
1 Rinki
1 Tamanna
1 Tishu
vboxuser@tamanna:~/Labfiles$ cat students.txt | wc -l
7
vboxuser@tamanna:~/Labfiles$ cat students.txt | wc -l
7
vboxuser@tamanna:~/Labfiles$
vboxuser@tamanna:~/Labfiles$ cat students.txt | tree report.txt
report.txt [error opening dir]

0 directories, 0 files
vboxuser@tamanna:~/Labfiles$ cat students.txt | tree report.txt
report.txt [error opening dir]

0 directories, 0 files
vboxuser@tamanna:~/Labfiles$ cat students.txt | tee report.txt
Tamanna
Rinki
Geetanjali
Tishu
Chikki
Enu
Mihru
vboxuser@tamanna:~/Labfiles$ cat students.txt | sort | tee report.txt
Chikki
Enu
Geetanjali
Mihru
Rinki
Tamanna
Tishu
vboxuser@tamanna:~/Labfiles$ cat abc.txt 2> error.txt
vboxuser@tamanna:~/Labfiles$ cat abc.txt 2>> error.txt
vboxuser@tamanna:~/Labfiles$ ls students.txt abc.txt >
-bash: syntax error near unexpected token `newline'
vboxuser@tamanna:~/Labfiles$ ls students.txt abc.txt > result.txt 2>&1
vboxuser@tamanna:~/Labfiles$ cat students.txt | sort | uniq -c | tee filtered_report.txt
1 Chikki
1 Enu
1 Geetanjali
1 Mihru
1 Rinki
1 Tamanna
1 Tishu
vboxuser@tamanna:~/Labfiles$

```

```

vboxuser@tamanna:~$ wc -l < students.txt
7
vboxuser@tamanna:~$ cat students.txt | sort
Chikki
Enu
Geetanjali
Mihru
Rinki
Tamanna
Tishu
vboxuser@tamanna:~$ cat students.txt | grep "Chikki"
Chikki
vboxuser@tamanna:~$ cat students.txt | sort | uniq
Chikki
Enu
Geetanjali
Mihru
Rinki
Tamanna
Tishu
vboxuser@tamanna:~$ cat students.txt | sort | uniq -c
1 Chikki
1 Enu
1 Geetanjali
1 Mihru
1 Rinki
1 Tamanna
1 Tishu
vboxuser@tamanna:~$ cat students.txt | wc -l
7
vboxuser@tamanna:~$ cat students.txt | tee report.txt
Chikki
Tamanna
Tishu
Geetanjali
Rinki
Enu
Mihru
vboxuser@tamanna:~$ cat abc.txt 2> error.txt
vboxuser@tamanna:~$ cat abc.txt 2>> error.txt
vboxuser@tamanna:~$ ls students.txt abc.txt > result.txt 2>&1
vboxuser@tamanna:~$ cat students.txt | sort | uniq -c | tee filtered_report.txt
1 Chikki
1 Enu
1 Geetanjali
1 Mihru
1 Rinki
1 Tamanna
1 Tishu
vboxuser@tamanna:~$

```

3. Create a College directory hierarchy with subdirectories (CSE, ECE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.

```
vboxuser@tamanna:~$ mkdir college
mkdir: college: File exists
vboxuser@tamanna:~$ cd college
vboxuser@tamanna:~/college$ mkdir -p CSE/Sem1/{DSA,DBMS} CSE/Sem2/{OS,PYTHON} ECE/Sem1/{Digital Electronics} ECE/Sem2/{Signals,Communication} ME/Sem1/{Thermodynamics,Mechanics} Civil/Sem1/{Surveying,Structures}
vboxuser@tamanna:~/college$ pwd
/home/vboxuser/college
vboxuser@tamanna:~/college$ tree
.
├── Civil
│   └── Sem1
│       ├── Structures
│       └── Surveying
├── CSE
│   ├── Sem1
│   │   ├── DBMS
│   │   └── DSA
│   └── Sem2
│       ├── OS
│       └── PYTHON
├── ECE
│   ├── Sem1
│   │   └── {Digital
│   │       └── Electronics}
│   └── Sem2
│       ├── Communication
│       └── Signals
├── Electronics}
├── linux
│   └── commands
├── ME
│   └── Sem1
│       ├── Mechanics
│       └── Thermodynamics
└── 25 directories, 0 files
vboxuser@tamanna:~/college$ cd CSE
vboxuser@tamanna:~/college/CSE$ ls
Sem1 Sem2
vboxuser@tamanna:~/college/CSE$ cd Sem1
vboxuser@tamanna:~/college/CSE/Sem1$ ls
DBMS DSA
vboxuser@tamanna:~/college/CSE/Sem1$ cd ../../
vboxuser@tamanna:~/college$ mkdir Test
vboxuser@tamanna:~/college$ rmdir Test
vboxuser@tamanna:~/college$ _
```

4. Configure file permissions and ownership.

```
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Downloads
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 15:50 Labfiles
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Videos
drwxrwxr-x 8 vboxuser vboxuser 4096 Sep 16 16:38 college
-rw-rw-r-- 1 vboxuser vboxuser  80 Sep 16 16:21 error.txt
-rw-rw-r-- 1 vboxuser vboxuser 104 Sep 16 16:22 filtered_report.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep  9 16:51 krmu
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 13:59 my
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:57 my-first-website
-rw-rw-r-- 1 vboxuser vboxuser  54 Sep 16 16:16 output.txt
-rwxr-xr-x 1 vboxuser vboxuser  0 Sep 16 16:43 permission.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 09:49 projectfile
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:20 report.txt
-rw-rw-r-- 1 vboxuser vboxuser  68 Sep 16 16:21 result.txt
drwx----- 5 vboxuser vboxuser 4096 Sep  9 18:00 snap
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 15 14:54 student
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:14 students.txt
-rw-rw-r-- 1 vboxuser vboxuser 212 Sep 15 14:49 test.txt
vboxuser@tamanna:~$ chmod o-w permission.txt
vboxuser@tamanna:~$ ls -l
total 92
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Downloads
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 15:50 Labfiles
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Videos
drwxrwxr-x 8 vboxuser vboxuser 4096 Sep 16 16:38 college
-rw-rw-r-- 1 vboxuser vboxuser  80 Sep 16 16:21 error.txt
-rw-rw-r-- 1 vboxuser vboxuser 104 Sep 16 16:22 filtered_report.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep  9 16:51 krmu
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 13:59 my
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:57 my-first-website
-rw-rw-r-- 1 vboxuser vboxuser  54 Sep 16 16:16 output.txt
-rwxr-xr-x 1 vboxuser vboxuser  0 Sep 16 16:43 permission.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 09:49 projectfile
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:20 report.txt
-rw-rw-r-- 1 vboxuser vboxuser  68 Sep 16 16:21 result.txt
drwx----- 5 vboxuser vboxuser 4096 Sep  9 18:00 snap
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 15 14:54 student
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:14 students.txt
-rw-rw-r-- 1 vboxuser vboxuser 212 Sep 15 14:49 test.txt
vboxuser@tamanna:~$
```

```

drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Videos
drwxrwxr-x 8 vboxuser vboxuser 4096 Sep 16 16:38 college
-rw-rw-r-- 1 vboxuser vboxuser  80 Sep 16 16:21 error.txt
-rw-rw-r-- 1 vboxuser vboxuser 104 Sep 16 16:22 filtered_report.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep  9 16:51 krmu
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:59 mj
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:57 mj-first-website
-rw-rw-r-- 1 vboxuser vboxuser  54 Sep 16 16:16 output.txt
-rwxr-xr-x 1 vboxuser vboxuser  0 Sep 16 16:43 permission.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 08:49 projectfile
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:20 report.txt
-rw-rw-r-- 1 vboxuser vboxuser  68 Sep 16 16:21 result.txt
drwx----- 5 vboxuser vboxuser 4096 Sep  9 18:00 snap
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 15 14:54 student
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:14 students.txt
-rw-r--r-- 1 vboxuser vboxuser 212 Sep 15 14:49 test.txt
vboxuser@tamanna:~$ sudo chown root:root permission.txt
[sudo] password for vboxuser:
Sorry, try again.
[sudo] password for vboxuser:
vboxuser@tamanna:~$ ls -l
total 92
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Downloads
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 15:50 Labfiles
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Videos
drwxrwxr-x 8 vboxuser vboxuser 4096 Sep 16 16:38 college
-rw-rw-r-- 1 vboxuser vboxuser  80 Sep 16 16:21 error.txt
-rw-rw-r-- 1 vboxuser vboxuser 104 Sep 16 16:22 filtered_report.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep  9 16:51 krmu
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:59 mj
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:57 mj-first-website
-rw-rw-r-- 1 vboxuser vboxuser  54 Sep 16 16:16 output.txt
-rwxr-xr-x 1 root root  0 Sep 16 16:43 permission.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 08:49 projectfile
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:20 report.txt
-rw-rw-r-- 1 vboxuser vboxuser  68 Sep 16 16:21 result.txt
drwx----- 5 vboxuser vboxuser 4096 Sep  9 18:00 snap
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 15 14:54 student
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:14 students.txt
-rw-r--r-- 1 vboxuser vboxuser 212 Sep 15 14:49 test.txt
vboxuser@tamanna:~$

```

```

vboxuser@tamanna:~$ cd ~
vboxuser@tamanna:~$ touch permission.txt
vboxuser@tamanna:~$ ls -l permission.txt
-rwxrwxr-x 1 vboxuser vboxuser 0 Sep 16 16:43 permission.txt
vboxuser@tamanna:~$ chmod +x permission.txt
vboxuser@tamanna:~$ ls -l
total 92
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Downloads
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 15:50 Labfiles
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep  9 16:45 Videos
drwxrwxr-x 8 vboxuser vboxuser 4096 Sep 16 16:38 college
-rw-rw-r-- 1 vboxuser vboxuser  80 Sep 16 16:21 error.txt
-rw-rw-r-- 1 vboxuser vboxuser 104 Sep 16 16:22 filtered_report.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep  9 16:51 krmu
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:59 mj
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 16 13:57 mj-first-website
-rw-rw-r-- 1 vboxuser vboxuser  54 Sep 16 16:16 output.txt
-rwxrwxr-x 1 vboxuser vboxuser  0 Sep 16 16:43 permission.txt
drwxrwxr-x 3 vboxuser vboxuser 4096 Sep 16 08:49 projectfile
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:20 report.txt
-rw-rw-r-- 1 vboxuser vboxuser  68 Sep 16 16:21 result.txt
drwx----- 5 vboxuser vboxuser 4096 Sep  9 18:00 snap
drwxrwxr-x 2 vboxuser vboxuser 4096 Sep 15 14:54 student
-rw-rw-r-- 1 vboxuser vboxuser  48 Sep 16 16:14 students.txt
-rw-r--r-- 1 vboxuser vboxuser 212 Sep 15 14:49 test.txt
vboxuser@tamanna:~$ chmod 755 permission.txt

```

